

–Curriculum Vitae–

Atanu Baksi, PhD

Notre Dame, IN, USA

Post-Doctoral Research Fellow, Department of Chemical and Biomolecular Engineering
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Research Interests

Mechanistic understanding of chemical transformation pathways in non-thermal plasma (NTP)-assisted heterogeneous catalysis, with focus on olefin selectivity control.

Molecular-level investigation of liquid-liquid phase separation (LLPS) in polyelectrolyte complexes and proteins using coarse-grained MD simulations, emphasizing the role of charge density and molecular asymmetry in phase behavior and macromolecular crowding.

Work Experience

Postdoctoral Researcher with Prof. William F. Schneider

July 2025 - Present

Department of Chemical and Biomolecular Engineering,
University of Notre Dame, Notre Dame, IN, USA

Developing 0D plasma kinetics models for DBD propane/ethane conversion; identifying mechanisms controlling olefin selectivity. Leading mentorship of four STRAND undergraduates on plasma catalysis modeling

Postdoctoral Researcher with Prof. Jonathan K. Whitmer

May 2024 - June 2025

Department of Chemical and Biomolecular Engineering,
University of Notre Dame, Notre Dame, IN, USA

Postdoctoral Researcher with Prof. Gül H. Zerbe

Feb 2022 - April 2024

Department of Chemical and Biomolecular Engineering
University of Houston, Houston, Texas, USA

Education

Ph.D. in Science (Physics)

2015 - 2021

Jadavpur University, Kolkata, West Bengal, India

Masters in Science - Physics (Condensed Matter Physics)

2012 - 2014

University of Calcutta, Kolkata, West Bengal, India

Undergraduate - Physics (Honors), Chemistry, Maths

2008 - 2012

University of Calcutta, Kolkata, West Bengal, India

Selected Publications and Projects

1. Modeling-Driven Control of Olefin Selectivity in Nonthermal DBD Propane Plasma., **A. Baksi**[#], C. Benaragama, D. Haycock, Jason C. Hicks*, William F. Schneider* (In preparation)
2. Nonthermal-Plasma-Driven Ethane Dehydrogenation: Kinetic Modeling of Known Chemistry, and Future Opportunities., Denver Haycock, Hannah Frankovic, Russell Clarke, **A. Baksi**, Jason C. Hicks, William F. Schneider* (To be submitted, 2026)
3. Polyelectrolyte Length Asymmetry Influences the Phase Composition and Viscoelasticity of Polyelectrolyte Complexes., D. Iyer, H. Senbandith, **A. Baksi**, V. Huaco, R. Goh, L. Willey, Jonathan K. Whitmer*, Samanvaya Srivastava* (To be submitted, 2026)
4. Chemical Specificity, Not Universal Entropy, Governs Osmotic Stability of Polyelectrolyte Coacervates, **A. Baksi**[#], A. Odenheimer[#], M. Farshad, I. Knight, N. Nguyen, R. Kayle, Samanvaya Srivastava,* Jonathan K. Whitmer* (To be submitted, 2026)
5. An explicit solvent model of coacervate structure and thermodynamics., **K. Alonso**[#], **A. Baksi**[#], I. Knight, N. Nguyen, M. Farshad, Samanvaya Srivastava, Jonathan K. Whitmer*, **J. Chem. Phys.** 2026, 164, 024904
6. The Molecular Picture of the Local Environment in a Stable Model Coacervate., **A. Baksi**, H Zerbe, A Agrawal, A Karim, Gül H Zerbe*, **Commun. Chem.** 2024, 7, 222
7. Dynamical Anomaly of Aqueous Amphiphilic Solutions: Connection to Solution H-Bond Fluctuation Dynamics?, **A. Baksi**[#], R. Biswas*, **ACS Omega** 2022, 7, 13, 10970-10984
8. Does Confinement Modify Preferential Solvation and H-bond Fluctuation Dynamics? A Molecular Level Investigation Through Simulations of Bulk and Confined Three-Component Mixture., **A. Baksi**[#], R Biswas*, **J. Phys. Chem. B** 2020, 124, 11718-11729
9. Dynamic Susceptibility and Structural Heterogeneity of Large Reverse Micellar Water: An Examination of the Core-Shell Model via Probing the Layer-wise Features., **A. Baksi**[#], Pradip K. Ghorai,* and Ranjit Biswas*, **J. Phys. Chem. B** 2020, 124, 2848-2863

Skills

Computational Methods:

Plasma kinetics modeling: 0D and quasi-1D DBD simulations (Bolsig++, BOLOS, ZDPlaskin, ChemPlaskin)

Classical molecular dynamics: GROMACS, LAMMPS, NAMD

Microkinetic modeling: CHEMKIN, RMG, RMS; VASP for DFT calculations

Trajectory analysis and visualization: MDAnalysis, MDTraj, VMD, OVITO, TRAVIS, ASE, VESTA

HPC and parallel computing: SLURM job management, MPI

Machine learning and neural networks: PyTorch, TensorFlow

Programming and Software Development:

Languages: Python (NumPy, SciPy, Cantera), FORTRAN, C++, Bash scripting

Workflow automation and data analysis pipelines

Professional Competencies:

Research mentorship and pedagogical design

Scientific presentation and technical communication

Collaborative project management across interdisciplinary teams

Awards and Fellowships

1. Qualified for CSIR-NET (June 2014; <1% success rate); awarded five-year Ph.D. Fellowship
2. Best POSTER presentation award in BOSE-FEST held in S. N. Bose National Center for Basic Sciences, March-2016
3. Best POSTER presentation award in DICB conference held in IISC, Bangalore, February-2019
4. GRC Carl Storm International Diversity Award for attending GRC Liquids 2019 held in Holderness, New Hampshire, USA

Oral Presentations

1. Research presentation delivered in APS March Meeting 2026 (virtual mode) held at Denver, Colorado
2. Research presentation delivered in APS March Meeting 2025 held at Anaheim, California
3. Oral presentation delivered in 11th Annual Texas Soft Matter Meeting, held at University of Texas at Austin, August 2022
4. Invited Oral presentation delivered in 'GORDON Research Seminar (GRS) Liquids 2019' held in Holderness School, USA during 1 - 4 August 2019
5. Oral presentation delivered in 'BOSE FEST' at S. N. Bose National Centre for Basic Sciences, India during 8-10 February 2018

Teaching and Mentoring Experience

Undergraduate Research Mentorship (University of Notre Dame, Jan, 2026–Present)

STRAND Program mentees under Prof. William F. Schneider:

- **Bridget MacDonough:** Role of Diluent gases in non-thermal plasma assisted propane dehydrogenation
- **Cole Ceklosky:** Non-oxidative plasma assisted propane dehydrogenation mechanism analysis
- **Josh Urban:** BOLSIG+ electron energy distribution function (EEDF) calculations for DBD reactors
- **Kyra Kleiman:** Role of temperature and electric field on product selectivity in NTP assisted propane dehydrogenation

Graduate Student Mentorship (University of Notre Dame, 2025)

Co-mentored under Prof. Jonathan K. Whitmer:

- **Kayley Alonso:** Coarse-grained molecular dynamics of explicit solvent coacervate systems
- **Arthur Odenheimer:** Osmotic stability and charge-density effects in polyelectrolyte complexes

Reviewing Experience

- Reviewer for Physics of Fluids (AIP Publication), February 2026
- Reviewer for International Society for Computer Assisted Orthopaedic Surgery, February 2026
- Sub-reviewer for ACS Catalysis, February, 2026
- Reviewer for Journal of Solution Chemistry (Springer Nature), January 2025
- Reviewer for ChemistrySelect Journal (Wiley), November 2024
- Reviewer for ChemistrySelect Journal (Wiley), August 2023
- Sub-reviewer for Journal of Chemical Physics and Journal of Physical Chemistry Letters

References

1. Dr. William F. Schneider (Professor)

Department of Chemical and Biomolecular Engineering, University of Notre Dame, USA

Email: wschneider@nd.edu

2. Dr. Jonathan K. Whitmer (Associate Professor)

Department of Chemical and Biomolecular Engineering, University of Notre Dame, USA

Email: jwhitme1@nd.edu

3. Dr. Gül H. Zerze (Assistant Professor)

Department of Chemical and Biomolecular engineering, University of Houston, USA

Email: gzerze@central.uh.edu, h.gul.ozar@gmail.com

4. Dr. Ranjit Biswas (Sr. Professor)

Department of Chemical, Biological & Macromolecular Sciences,
S. N. Bose National Centre for Basic Sciences, India

Email: ranjit@bose.res.in, rbsnbc@gmail.com

5. Dr. Samanvaya Srivastava (Associate Professor)

Department of Chemical and Biomolecular engineering, UCLA, USA

Email: samsri@ucla.edu

6. Dr. Hemant K. Kashyap (Professor)

Department of Chemistry, IIT Delhi, India

Email: hkashyap@chemistry.iitd.ac.in

7. Dr. Pradip Kumar Ghorai (Professor)

Department of Chemical Sciences, IISER Kolkata, India

Email: pradip@iiserkol.ac.in